

March 5, 2025

Meeting Priorities

1. Submit Example Code to Eval Platform (Owen)
2. Create Flink Data Ingestion
3. Saturation Operator

JAVA Version Commands:

- 1) Get the pom.xml file which lays out the dependencies we need thus far for our java file.
- 2) Download all the dependencies using mvn clean install -U
- 3) Java file should be all good to go after. You can add an argument "<http://127.0.0.1:8866>" to IntelliJ and then click run and it should compile and execute the file.
- 4) Current work needs to be done in the Processing Function to decode the image data from our batch.

//ignore

Commands for Java Sample Solution when building manually:

- 1) javac -cp .:json-20250107.jar ClientRef.java
- 2) java -cp .:json-20250107.jar ClientRef <http://127.0.0.1:8866>
- 3) mvn clean install -U

March 6, 2025

Meeting w/ Professor

- Need to figure out what they mean by fault tolerance
 - High availability or high consistency?
- Start flink w extra task managers that do nothing and if one of them fails, flink will transfer job to another empty manager (this addresses availability)
- Outlier Detection per point within each tile, don't need to span tiles

Goals to be Completed by March 19th

Owen

- Have submitted naive solution
- Document how to do so
- Help Custom Window Logic

Tanish:

- Image deserialization
- Process Function
 - Saturation
 - Outliers
 - Clustering per tile
 - DBSCAN

Pmesh

- Help Image deserialization
- Custom Window Logic

Optimization Ideas

- Parallelize saturation analysis
- MinHeap